

MONOTHERAPY

Adverse Reactions Reported with IRINOL in Monotherapy (350 mg/m ² every 3 weeks schedule)		
MedDRA System Organ Class	Frequency Category	Preferred Term
Infections and infestations	Common	Infection
Blood and lymphatic system disorders	Very Common	Neutropenia
	Very Common	Anaemia
	Common	Thrombocytopenia
	Common	Febrile neutropenia
Metabolism and nutrition disorders	Very Common	Decreased appetite
Nervous system disorders		Cholinergic syndrome
Gastrointestinal disorders	Very Common	Diarrhoea, vomiting, nausea, abdominal pain,constipation
Skin and subcutaneous tissue disorders	Very Common	Alopecia (reversible)
General disorders and administration site conditions	Very Common	Mucosal Inflammation, pyrexia, asthenia
Investigations	Common	Blood creatinine increased, transaminases (SGPT and SGOT) increased, blood alkaline phosphatase increased

Laboratory tests: Transient and mild to moderate increases in serum levels of either transaminases, alkaline phosphatase or bilirubin were observed in the patients, respectively, in the absence of progressive liver metastasis. Transient and mild to moderate increases of serum levels of creatinine have been observed in the patients.

COMBINATION THERAPY
Adverse reactions detailed in this section refer to irinotecan. There is no evidence that the safety profile of irinotecan is influenced by cetuximab or vice versa. In combination with cetuximab, additional reported adverse reactions were those expected with cetuximab, such as acneiform rash. For information on adverse reactions to irinotecan in combination with cetuximab, also refer to their respective summary of product characteristics. Adverse drug reactions reported in patients treated with capecitabine in combination with irinotecan in addition to those seen with capecitabine monotherapy or seen at a higher frequency grouping compared to capecitabine monotherapy include:
Very common, all grade adverse drug reactions: thrombosis/embolism; Common, all grade adverse drug reactions: hypersensitivity reaction, cardiac ischemia/infarction; Common, grade 3 and grade 4 adverse drug reactions: febrile neutropenia. For complete information on adverse reactions of capecitabine, refer to the capecitabine summary product of characteristics. Grade 3 and Grade 4 adverse drug reactions reported in patients treated with capecitabine in combination with irinotecan and bevacizumab in addition to those seen with capecitabine monotherapy or seen at a higher frequency grouping compared to capecitabine monotherapy include:
Common, grade 3 and grade 4 adverse drug reactions: neutropenia, thrombosis/embolism, hypertension, and cardiac ischemia/infarction. For complete information on adverse reactions of capecitabine and bevacizumab, refer to the respective capecitabine and bevacizumab summary of product characteristics. Grade 3 hypertension was the principal significant risk involved with the addition of bevacizumab to bolus IRINOL/5-FU/FA. In addition, there was a small increase in the grade 3/4 chemotherapy adverse events of diarrhoea and leukopenia with this regimen compared to patients receiving bolus IRINOL/5-FU/FA alone. For other information on adverse reactions in combination with bevacizumab, refer to the bevacizumab summary of product characteristics. Safety data of adverse reactions from clinical studies demonstrate very commonly observed NCI Grade 3 or 4 possibly or probably-related adverse events in the blood and the lymphatic system disorders, gastrointestinal disorders, and skin and subcutaneous tissue disorders MedDRA System Organ Classes. The following adverse reactions considered to be possibly or probably related to the administration of IRINOL have been reported from 145 patients treated by IRINOL in combination therapy with 5FU/FA in every 2 weeks schedule at the recommended dose of 180 mg/m².

Adverse Reactions Reported with IRINOL in Monotherapy (180 mg/m ² every 2 weeks schedule)		
MedDRA System Organ Class	Frequency Category	Preferred Term
Infections and infestations	Common	Infection
Blood and lymphatic system disorders	Very Common	Neutropenia, Anaemia, Thrombocytopenia
	Common	Febrile neutropenia
Metabolism and nutrition disorders	Very Common	Decreased appetite
Nervous system disorders	Very Common	Cholinergic syndrome
Gastrointestinal disorders	Very Common	Diarrhoea, Vomiting, Nausea
	Common	Abdominal pain, Constipation
Skin and subcutaneous tissue disorders	Very Common	Alopecia (reversible)
General disorders and administration site conditions	Very Common	Mucosal Inflammation, Asthenia
	Common	Pyrexia
Investigations	Very Common	Blood creatinine increased, transaminases (SGPT and SGOT) increased, blood alkaline phosphatase increased

Laboratory tests: Transient serum levels (grades 1 and 2) of either SGPT, SGOT, alkaline phosphatase or bilirubin were observed in the patients, respectively, in the absence of progressive liver metastasis. Transient grade 3 were observed in the patients, respectively. No grade 4 was observed. Increases of amylase and/or lipase have been very rarely reported. Rare cases of hypokalaemia and hyponatremia mostly related with diarrhoea and vomiting have been reported.

POST-MARKETING SURVEILLANCE
Side effects have been summarised in the table below with MedDRA frequencies.

MedDRA System Organ Class	Adverse Reaction
Infections and infestations	<i>Pseudomonas</i> colitis one of which has been documented bacteriologically (<i>Clostridium difficile</i>) - Sepsis
Blood and lymphatic systemic disorders	Peripheral thrombocytopenia with antiplatelet antibodies

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Metabolism and nutrition disorders	- Dehydration (due to diarrhoea and vomiting) - Hypovolemia
Immune system disorders	- Hypersensitivity reaction - Anaphylactic reaction
Nervous system disorders	- Speech disorders generally transient in nature,in some cases, the event was attributed to thecholinergic syndrome observed during orshortly after infusion of Irinotecan - Parosethisia
Cardiac disorders	- Hypertension (during or after infusion) - Cardio circulatory failure*
Respiratory, thoracic and mediastinal disorder	- Interstitial pulmonary disease presenting aspulmonary infiltrates is uncommon duringirinotecan therapy - Dyspnoea - Hiccups
Gastrointestinal disorders	- Intestinal obstruction - Ileus: cases of ileus without preceding colitis have also been reported - Megacolon - Gastrointestinal haemorrhage - Colitis; in some cases, colitis was complicated by ulceration, bleeding, ileus, or infection - Typhlitis - Ischemic colitis - Ulcerative colitis - Gastrointestinal bleeding - Symptomatic or asymptomatic elevated pancreatic enzymes - Intestinal perforation
Skin and subcutaneous tissue disorders	Skin reactions
General disorder and administration site conditions	Infusion site reactions
Investigations	- Blood amylase increased - Lipase increased - Hypokalaemia - Hyponatremia mostly related with diarrhoea and vomiting - Increases in serum levels of transaminases (i.e. AST and ALT) in the absence of progressive liver metastasis have been very rarely reported
Musculoskeletal and connective tissue disorders	Muscular contraction or cramps
Renal and urinary disorders	- Renal impairment and acute renal failure generally in patients who become infected and/or volume depleted from severe gastrointestinal toxicities* - Renal insufficiency*
Vascular disorders	Hypotension*

* Infrequent cases of renal insufficiency, hypotension or cardio circulatory failure have been observed in patients who experienced episodes of dehydration associated with diarrhoea and/or vomiting, or sepsis.

SYMPTOMS AND TREATMENT OF OVERDOSE
There have been reports of overdosage at doses up to approximately twice the recommended therapeutic dose, which may be fatal. The most significant adverse reactions reported were severe neutropenia and severe diarrhoea. There is no known antidote for IRINOL. Maximum supportive care should be instituted to prevent dehydration due to diarrhoea and to treat any infectious complications.

EFFECTS ON ABILITY TO DRIVE AND USE MACHINE
Patients should be warned about the potential for dizziness or visual disturbances which may occur within 24 hours following the administration of IRINOL, and advised not to drive or operate machinery if these symptoms occur.

INSTRUCTION FOR USE
As with other antineoplastic agents, IRINOL must be prepared and handled with caution. The use of glasses, mask and gloves is required. If IRINOL solution or infusion solution should come into contact with the skin, wash immediately and thoroughly with soap and water. If IRINOL solution or infusion solution should come into contact with the mucous membranes, wash immediately with water. **Preparation for the intravenous infusion administration:** As with any other injectable medicinal products, the IRINOL solution must be prepared aseptically. If any precipitate is observed in the vials or after dilution, the product should be discarded according to standard procedures for cytotoxic agents. Aseptically withdraw the required amount of IRINOL solution from the vial with a calibrated syringe and inject into a 250 ml infusion bag or bottle containing either 0.9% sodium chloride solution or 5% glucose solution. The infusion should then be thoroughly mixed by manual rotation. The final concentration of Irinol (Irinotecan Concentrate for Solution For Infusion 20mg/ml) after dilution according to the above information is between 0.12 mg/ml to 3.0 mg/ml. **Incompatibilities:** None known. In the absence of compatibility studies, this medicinal product must not be mixed with other medicinal products. **Disposal:** Any unused medicinal product or waste material should be disposed of in accordance with local requirements.

STORAGE CONDITION
Unopened vial: Store below 30°C and store the vial in the original packaging to protect from light. **After further dilution:** The solution is physically and chemically stable with infusion solutions; 0.9% Sodium Chloride Injection and 5% Dextrose Injection for up to 24 hours at room temperature (below 30°C) and up to 48 hours when stored at 2°C -8°C. It is recommended that to reduce the microbiological hazard, the infusion solutions should be prepared immediately before use and infusion commenced as soon as practicable after preparation. If not used immediately, in-use storage times and conditions prior to use are the responsibility of the user and should not be longer than 24 hours at 2°C to 8°C, unless dilution has taken place in controlled and validated aseptic conditions.

SHELF LIFE
Unopened vial: The shelf-life of unopened vials is 24 months.

After further dilution: The solution is physically and chemically stable with infusion solutions; 0.9% Sodium Chloride Injection and 5% Dextrose Injection for up to 24 hours at room temperature (below 30°C) and up to 48 hours when stored at 2°C -8°C. It is recommended that to reduce the microbiological hazard, the infusion solutions should be prepared immediately before use and infusion commenced as soon as practicable after preparation. If not used immediately, in-use storage times and conditions prior to use are the

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responsibility of the user and should not be longer than 24 hours at 2°C to 8°C, unless dilution has taken place in controlled and validated aseptic conditions.

PRESENTATION
IRINOL (Irinotecan 40mg/2ml Concentrate for Solution For Infusion) is packed in Amber color glass vials (type I) with 4ml nominal capacity, with reinforced non-PVC base, which are stoppered with grey butyl rubber plugs, creamped with flip off seal grey colour with non-tear off aluminium seals and tampered with biogreen flip - off - tops. Each ml of the solution contains 20 mg of Irinotecan Hydrochloride Trihydrate of the active principle.

IRINOL (Irinotecan 100mg/5ml Concentrate For Solution For Infusion) is packed in Amber color glass vials (type I) with 5ml nominal capacity, with reinforced non-PVC base, which are stoppered with grey butyl rubber plugs, creamped with flip off seal grey colour with non-tear off aluminium seals and tampered with biogreen flip - off – tops. Each ml of the solution contains 20 mg of Irinotecan Hydrochloride Trihydrate of the active principle.

Product Registration Holder:
Unimed Sdn Bhd
53, Jalan Tembunga SD 5/2B, Bandar Sri Damansara, 52200, Kuala Lumpur, Malaysia

Manufactured by:
VENUS REMEDIES LIMITED
Hill Top Industrial Estate, Jhuangjari, EPIP, Phase-I (Extn.), Bhatoli Kalan, Baddi, Distt. Solan, Himachal Pradesh-173205 (INDIA)

Date of Revision: 9th March 2021

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IRINOL

Irinotecan Concentrate for Solution for Infusion 20 mg/ml

For IV Infusion

PRODUCT DESCRIPTION
Description of finished product (before dilution): Clear colorless to light pale yellow solution, free from visible extraneous particulate matter.
Description after Dilution: In 0.9% Sodium Chloride Injection: Clear colorless to light pale yellow solution, free from visible extraneous particulate matter.
In 5% glucose Injection: Clear colorless to light pale yellow solution, free from visible extraneous particulate matter.

PHARMACODYNAMICS
Cytostatic topoisomerase I inhibitor.
ATC Code: L01XX19
Irinotecan is a semi-synthetic derivative of camptothecin. It is an antineoplastic agent which acts as a specific inhibitor of DNA topoisomerase I. It is metabolised by carboxylesterase in most tissues to SN-38, which was found to be more active than irinotecan in purified topoisomerase I and more cytotoxic than irinotecan against several murine and human tumour cell lines. The inhibition of DNA topoisomerase I by irinotecan or SN-38 induces single-strand DNA lesions which blocks the DNA replication fork and are responsible for the cytotoxicity. This cytotoxic activity was found time-dependent and was specific to the S phase. In vitro, irinotecan and SN-38 were not found to be significantly recognised by the P - glycoprotein MDR, and displays cytotoxic activities against doxorubicin and vinblastine resistant cell lines. Furthermore, irinotecan has a broad antitumor activity in vivo against murine tumour models (PD3 pancreatic ductal adenocarcinoma, MA16C mammary adenocarcinoma, C38 and C51 colon adenocarcinomas) and against human xenografts (Co-4 colon adenocarcinoma, Mx-1 mammary adenocarcinoma, ST-15 and SC-16 gastric adenocarcinomas). Irinotecan is also active against tumors expressing the P-glycoprotein MDR (vincristine- and doxorubicin-resistant P388 leukaemia's). Beside the antitumor activity of irinotecan, the most relevant pharmacological effect of irinotecan is the inhibition of acetylcholinesterase. **In combination therapy for the first-line treatment of metastatic colorectal carcinoma.**
In combination therapy with Folinic Acid and 5-Fluorouracil
In the every 2 weeks schedule, on day 1, the administration of IRINOL at 180 mg/m² once every 2 weeks is followed by infusion with folinic acid (200 mg/m² over a 2-hour intravenous infusion) and 5-fluorouracil (400 mg/m² as an intravenous bolus, followed by 600 mg/m² over a 22-hour intravenous infusion). On day 2, folinic acid and 5-fluorouracil are administered at the same doses and schedules. In the weekly schedule, the incidence of severe diarrhoea in patients treated by IRINOL in combination with 5FU/FA and in patients treated by 5FU/FA alone. The incidence of severe neutropenia (neutrophil count < 500 cells/mm³) was in patients treated by IRINOL in combination with 5FU/FA and in patients treated by 5FU/FA alone. **In combination therapy with bevacizumab:** Bevacizumab used in combination with IRINOL/5FU/FA as first-line treatment for metastatic carcinoma of the colon or rectum. The addition of bevacizumab to the combination of IRINOL/5FU/FA resulted in a statistically significant increase in overall survival. The clinical benefit, as measured by overall survival, was seen in all patient subgroups, including those defined by age, sex, and performance status, location of primary tumour, number of organs involved, and duration of metastatic disease. **In combination therapy with capecitabine:** The support the use of capecitabine at a starting dose of 1000 mg/m² for 2 weeks every 3 weeks in combination with irinotecan for the first-line treatment of patients with metastatic colorectal cancer. Sequential treatment consisted of first-line treatment with capecitabine (1250 mg/m² twice daily for 14 days), second-line irinotecan (350 mg/m² on day 1), and third-line combination of capecitabine (1000 mg/m² twice daily for 14 days) with oxaliplatin (130 mg/m² on day 1). Combination treatment consisted of first-line treatment of capecitabine (1000 mg/m² twice daily for 14 days) combined with irinotecan (250 mg/m² on day 1) and second-line capecitabine (1000 mg/m² twice daily for 14 days) plus oxaliplatin (130 mg/m² on day 1). All treatment cycles were administered at intervals of 3 weeks. In first-line treatment the median progression-free survival in the intent-to-treat population was 5.8 months for capecitabine monotherapy and 7.8 months for irinotecan. The intensity of the major toxicities encountered with IRINOL (e.g., leukopenutropenia and diarrhoea) are related to the exposure (AUC) to parent drug and metabolite SN-38. Significant correlations were observed between haematological toxicity (decrease in white blood cells and neutrophils at nadir) or diarrhoea intensity and both irinotecan and metabolite SN-38 AUC values in monotherapy. **Patients with Reduced UGT1A1 Activity:** Uridine diphosphate-glucuronosyl transferase 1A1 (UGT1A1) is involved in the metabolic deactivation of SN-38, the active metabolite of irinotecan to inactive SN-38 glucuronide (SN-38G). The UGT1A1 gene is highly polymorphic, resulting in variable metabolic capacities among individuals. One specific variation of the UGT1A1 gene includes a polymorphism in the promoter region known as the UGT1A1*28 variant. This variant indicate that individuals with Crigler-Najjar syndrome (types 1 and 2) or those who are homozygous for the UGT1A1*28 allele (Gilbert's syndrome) are at increased risk of haematological toxicity (grades 3 and 4) following administration of irinotecan at moderate or high doses (>150 mg/m²). A relationship between UGT1A1 genotype and the occurrence of irinotecan induced diarrhea was not established. Patients known to be homozygous for UGT1A1*28 should be administered the normally indicated irinotecan starting dose. However, these patients should be monitored for haematologic toxicities. A reduced irinotecan starting dose should be considered for patients who have experienced prior hematologic toxicity with previous treatment. The exact reduction in starting dose in this patient population has not been established and any subsequent dose modifications should be based on a patient's tolerance of the treatment. There is at present insufficient data to conclude on clinical utility of UGT1A1 genotyping.

PHARMACOKINETICS
Absorption: At the end of the infusion, at the recommended dose of 350 mg/m², the mean peak plasma concentrations of irinotecan and SN-38 were 7.7 µg/ml and 56 ng/ml, respectively, and the mean area under the curve (AUC) values were 34 µg.h/ml and 451 ng.h/ml, respectively. A large interindividual variability in pharmacokinetic parameters is generally observed for SN-38. **Distribution:** In vitro, plasma protein binding for irinotecan and SN-38 was approximately 65% and 95%, respectively. **Biotransformation:** Mass balance and metabolism studies with 14 C-labelled drug have shown

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that more than 50% of an intravenously administered dose of irinotecan is excreted as unchanged drug, with 33% in the faeces mainly via the bile and 22% in urine. Two metabolic pathways account each for at least 12% of the dose: - Hydrolysis by carboxylesterase into active metabolite SN-38, SN-38 is mainly eliminated by glucuronidation, and further by biliary and renal excretion (less than 0.5% of the irinotecan dose) - The SN-38 glucuronide is subsequently probably hydrolysed in the intestine. - Cytochrome P450 3A enzymes-dependent oxidations resulting in opening of the outer piperidine ring with formation of APC (aminopentanoic acid derivate) and NPC (primary amine derivate). - Unchanged irinotecan is the major entity in plasma, followed by APC, SN-38 glucuronide and SN-38. Only SN-38 has significant cytotoxic activity. Elimination: Irinotecan showed a biphasic or triphasic elimination profile. The mean plasma clearance was 15 L/h/m² and the volume of distribution at steady state (V_{ss}): 157 L/m². The mean plasma half-life of the first phase of the triphasic model was 12 minutes, of the second phase 2.5 hours, and the terminal phase half-life was 14.2 hours. SN-38 showed a biphasic elimination profile with a mean terminal elimination half-life of 13.8 hours. Irinotecan clearance is decreased by about 40% in patients with bilirubinemia between 1.5 and 3 times the upper normal limit. In these patients a 200 mg/m² irinotecan dose leads to plasma drug exposure comparable to that observed at 350 mg/m² in cancer patients with normal liver parameters. Linearity/non-linearity: A population pharmacokinetic analysis of irinotecan has been performed in patients with metastatic colorectal cancer, treated with various schedules and at different doses. All studies have shown that irinotecan (CPT-11) and SN-38 exposure increase proportionally with CPT-11 administered dose; their pharmacokinetics are independent of the number of previous cycles and of the administration schedule. INDICATION IRINOL (Irinotecan Concentrate For Solution For Infusion 20 mg/ml) is indicated for the treatment of patients with advanced colorectal cancer: - In combination with 5-Fluorouracil and folinic acid in patients without prior chemotherapy for advanced disease. - As a single agent in patients who have failed an established 5-Fluorouracil containing treatment regimen. RECOMMENDED DOSE Posology and method of administration: Strictly follow the recommended dosage unless directed otherwise by the physician. All doses of irinotecan should be administered as an intravenous infusion over 30 to 90 minutes. Single-agent dosage schedules: Single-agent dosage schedules have been extensively studied for metastatic colorectal cancer. Starting dose Weekly Dosage Schedule: The recommended single-agent starting dose of irinotecan is 125 mg/m². A lower starting dose may be considered (e.g., 100 mg/m²) for patients with any of the following conditions: prior extensive radiotherapy, performance status of 2, increased bilirubin levels, or gastric cancer. Treatment should be given in repeated 6-week cycles, comprising weekly treatment for 4 weeks, followed by a 2-week rest. Once-Every-2-Week Dosage Schedule: The usual recommended starting dose of irinotecan is 250 mg/m² every 2 weeks by intravenous infusion. A lower starting dose may be considered (e.g., 200 mg/m²) for patients with any of the following conditions: age 65 years and older, prior extensive radiotherapy, performance status of 2, increased bilirubin levels, or gastric cancer. Once-Every-3-Week Dosage Schedule: The usual recommended starting dose of irinotecan for the once-every-3-week dosage schedule is 350 mg/m². A lower starting dose may be considered (e.g., 300 mg/m²) for patients with any of the following conditions: age 65 years and older, prior extensive radiotherapy, performance status of 2, increased bilirubin levels, or gastric cancer. SPECIAL POPULATIONS Elderly: The dose should be chosen carefully in this population due to their greater frequency of decreased biological functions. This population should require more intensive surveillance (See section Pharmacokinetic Properties). Patients with impaired hepatic function: In patients with hepatic dysfunction, the following starting doses are recommended: <table><tr><th>Serum Total Bilirubin Concentration</th><th>Serum ALT/AST Concentration</th><th>Starting Dose, mg/m²</th></tr><tr><td>1.5-3.0 x IULN</td><td>≤5.0 x IULN</td><td>60</td></tr><tr><td>3.1-5.0 x IULN</td><td>≤5.0 x IULN</td><td>50</td></tr><tr><td><1.5 x IULN</td><td>5.1-20.0 x IULN</td><td>60</td></tr><tr><td>1.5-5.0 x IULN</td><td>5.1-20.0 x IULN</td><td>40</td></tr></table> Table 2: Starting Doses in Patients with Hepatic Dysfunction: <i>Single-agent Once-Every-3-Week Regimen</i> <table><tr><th>Serum Total Bilirubin Concentration</th><th>Starting Dose, mg/m²</th></tr><tr><td>1.5-3.0 x IULN</td><td>200</td></tr><tr><td>>3.0 x IULN</td><td>Not Recommended^a</td></tr></table> ^a The safety and pharmacokinetics of irinotecan given once-every-3-weeks have not been defined in patients with bilirubin >3.0 x institutional upper limit of normal (IULN) and this schedule cannot be recommended in these patients. Patients with impaired renal function: Studies in this population have not been conducted (See section Pharmacokinetic Properties). Therefore, caution should be undertaken in patients with impaired renal function. Irinotecan is not recommended for use in patients on dialysis. COMBINATION-AGENT DOSAGE SCHEDULES Starting Dose: Irinotecan in Combination with 5-Fluorouracil (5-FU) and folinic acid (FA). Irinotecan in combination with 5-FU and folinic acid is recommended for use in patients with metastatic colorectal cancer. For all regimens, the dose of folinic acid should be administered immediately after irinotecan, with the administration of 5-FU to occur immediately after receipt of folinic acid. The currently recommended regimens are shown below: Regimen 1 (6-week cycle with bolus 5-FU/FA): The recommended starting dose is 125 mg/m² of irinotecan, 500 mg/m² bolus 5-FU, and 20 mg/m² bolus folinic acid. Regimen 2 (6-week cycle with infusional 5-FU/FA): The recommended starting dose is 180 mg/m² of irinotecan, 400 mg/m² bolus 5-FU, 600 mg/m² 5-FU infusion, and 200 mg/m² folinic acid. Lower starting doses may be considered for irinotecan (e.g., 100 mg/m²) and 5-FU (e.g., 400 mg/m²) for patients with any of the following conditions: age 65 years and older, prior extensive radiotherapy, performance status of 2, increased bilirubin levels, or gastric cancer. Treatment should be given in repeated 6-week cycles, comprising weekly treatment for 4 weeks, followed by a 2-week rest. Duration of treatment: For both single-agent and combination-agent regimens, treatment with additional cycles of irinotecan may be continued indefinitely in patients who attain a tumor response or in patients whose cancer remains stable. Patients should be carefully monitored for toxicity and should be removed from therapy if unacceptable toxicity occurs that is not responsive to dose modification and routine supportive care. Dose modification recommendations: The recommended dose modifications during a cycle of therapy and at the start of each subsequent cycle of therapy for single-agent dosage schedules are described in Table 3. These recommendations are based on toxicities commonly observed with the administration of irinotecan. For modifications at the start of a subsequent cycle of therapy,		Serum Total Bilirubin Concentration	Serum ALT/AST Concentration	Starting Dose, mg/m ²	1.5-3.0 x IULN	≤5.0 x IULN	60	3.1-5.0 x IULN	≤5.0 x IULN	50	<1.5 x IULN	5.1-20.0 x IULN	60	1.5-5.0 x IULN	5.1-20.0 x IULN	40	Serum Total Bilirubin Concentration	Starting Dose, mg/m ²	1.5-3.0 x IULN	200	>3.0 x IULN	Not Recommended ^a	the dose of irinotecan should be decreased relative to the initial dose of the previous cycle. The recommended dose modifications during a cycle of therapy and at the start of each subsequent cycle of therapy for irinotecan, 5-FU, and folinic acid are described in Table 4. All dose modifications should be based on the worst preceding toxicity. A new cycle of therapy should not begin until the toxicity has recovered to Grade 2 or less. Treatment may be delayed 1 to 2 weeks to allow for recovery from treatment-related toxicity. If the patient has not recovered, consideration should be given to discontinuing irinotecan. Table 3: Recommended Dose Modifications for Single-agent Schedules A new cycle of therapy should not begin until the granulocyte count has recovered to ≥1500/mm³, and the platelet count has recovered to ≥100,000/mm³, and treatment-related diarrhea is fully resolved. Treatment should be delayed 1 to 2 weeks to allow for recovery from treatment-related toxicities. If the patient has not recovered after a 2-week delay, consideration should be given to discontinuing irinotecan. <table><tr><th>Toxicity NCI Grade^a (Value)</th><th>During a Cycle of Therapy</th><th>At the Start of the Next Cycle of Therapy (After Adequate Recovery), Compared with the Starting Dose in the Previous Cycle^b</th></tr><tr><th></th><th>Weekly</th><th>Weekly</th><th>Once every 2 or 3 Week</th></tr><tr><td>No toxicity</td><td>Maintain dose level</td><td>↓ 25 mg/m² up to a maximum dose of 150 mg/m²</td><td>Maintain dose level</td></tr><tr><td>Neutropenia 1 (1500 to 1999/mm³) 2 (1000 to 1499/mm³) 3 (500 to 999/mm³) 4 (<500/mm³)</td><td>Maintain dose level ↓ 25 mg/m² Omit dose, then ↓ 25 mg/m² when resolved to ≤ Grade 2</td><td>Maintain dose level ↓ 25 mg/m² Maintain dose level ↓ 25 mg/m² when resolved to ≤ Grade 2</td><td>Maintain dose level ↓ 50 mg/m² ↓ 50 mg/m²</td></tr><tr><td>Neutropenic fever (Grade 4 neutropenia & ≥ Grade 2 fever)</td><td>Omit dose then ↓ 50 mg/m² when resolved</td><td>↓ 50 mg/m²</td><td>↓ 50 mg/m²</td></tr><tr><td>Other hematologic toxicities</td><td colspan="3">Dose modifications for leukopenia, thrombocytopenia, and anaemia during a cycle of therapy and at the start of subsequent cycles of therapy are also based on NCI toxicity criteria and are the same as recommended for neutropenia above</td></tr><tr><td>Diarrhea 1 (2-3 stools/day >pretx) 2 (4-6 stools/day >pretx) 3 (7-9 stools/day >pretx) 4 (≥ 10 stools/day >pretx)</td><td>Maintain dose level ↓ 25 mg/m² Omit dose, then ↓ 25 mg/m² when resolved to ≤ Grade 2</td><td>Maintain dose level ↓ 25 mg/m² ↓ 50 mg/m² when resolved to ≤ Grade 2</td><td>Maintain dose level ↓ 50 mg/m² ↓ 50 mg/m²</td></tr><tr><td>Other non-hematologic toxicities^c 1 2 3 4</td><td>Maintain dose level ↓ 25 mg/m² Omit dose, then ↓ 25 mg/m² when resolved to ≤ Grade 2</td><td>Maintain dose level ↓ 25 mg/m² ↓ 50 mg/m² when resolved to ≤ Grade 2</td><td>Maintain dose level ↓ 50 mg/m² ↓ 50 mg/m²</td></tr></table> ^a All dose modifications should be based on the worst preceding toxicity. ^b National Cancer Institute Common Toxicity Criteria. ^c Pre-treatment. ^d Excludes alopecia, anorexia, asthenia. Table 4: Recommended Dose Modifications for Irinotecan-5-Fluorouracil/Folinic Acid Combination Schedules Patients should return to pre-treatment bowel function without requiring anti-diarrhea medications for at least 24 hours before the next chemotherapy administration. A new cycle of therapy should not begin until the granulocyte count has recovered to ≥1500/mm³, and the platelet count has recovered to ≥100,000/mm³, and treatment-related diarrhea is fully resolved. Treatment should be delayed 1 to 2 weeks to allow for recovery from treatment-related toxicities. 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For additional contraindications of cetuximab or bevacizumab or capecitabine, refer to the product information for these medicinal products. WARNINGS AND PRECAUTIONS The use of IRINOL should be confined to units specialised in the administration of cytotoxic chemotherapy and it should only be administered under the supervision of a physician qualified in the use of anticancer chemotherapy. Given the nature and incidence of adverse events, IRINOL will only be prescribed in the following cases after the expected benefits have been weighted against the possible therapeutic risks: - In patients presenting a risk factor, particularly those with a WHO performance status = 2. - In the few rare instances where patients are deemed unlikely to observe recommendations regarding management of adverse events (need for immediate and prolonged anti-diarrhoeal treatment combined with high fluid intake at onset of delayed diarrhoea). Strict hospital supervision is recommended for such patients. When IRINOL is used in monotherapy, it is usually prescribed with the every-3-week-dosage schedule. However, the weekly-dosage schedule may be considered in patients who may need a closer follow-up or who are at particular risk of severe neutropenia. Delayed diarrhoea: Patients should be made aware of the risk of delayed diarrhoea occurring more than 24 hours after the administration of IRINOL and at any time before the next cycle. In monotherapy, the median time of onset of the first liquid stool was on day 5 after the infusion of IRINOL. Patients should quickly inform their physician of its occurrence and start appropriate therapy immediately. Patients with an increased risk of diarrhoea are those who had a previous abdominal and/or pelvic radiotherapy, those with baseline hyperleucocytosis, those with performance status ≥ 2 and women. If not properly treated, diarrhoea can be life-threatening, especially if the patient is concomitantly neutropenic. As soon as the first liquid stool occurs, the patient should start drinking large volumes of beverages containing electrolytes and an appropriate anti-diarrhoeal therapy must be initiated immediately. This anti-diarrhoeal treatment will be prescribed by the department where IRINOL has been administered. After discharge from the hospital, the patients should obtain the prescribed drugs so that they can treat the diarrhoea as soon as it occurs. In addition, they must inform their physician or the department administering IRINOL when and or if diarrhoea is occurring. The currently recommended anti-diarrhoeal treatment consists of high doses of loperamide (4 mg for the first intake and then 2 mg every 2 hours). This therapy should continue for 12 hours after the last liquid stool and should not be modified. In no instance should loperamide be administered for more than 48 consecutive hours at these doses, because of the risk of paralytic ileus, nor for less than 12 hours. 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In patients who experienced an acute and severe cholinergic syndrome, the use of prophylactic atropine sulphate is recommended with subsequent doses of IRINOL. Respiratory disorders: Interstitial pulmonary disease presenting as pulmonary infiltrates is uncommon during irinotecan therapy. Interstitial pulmonary disease can be fatal. Risk factors possibly associated with the development of interstitial pulmonary disease include the use of pneumotoxic drugs, radiation therapy and colony stimulating factors. Patients with risk factors should be closely monitored for respiratory symptoms before and during irinotecan therapy. Extravasation: While irinotecan is not a known vesicant, care should be taken to avoid extravasation and the infusion site should be monitored for signs of inflammation. Should extravasation occur, flushing the site and application of ice is recommended. 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